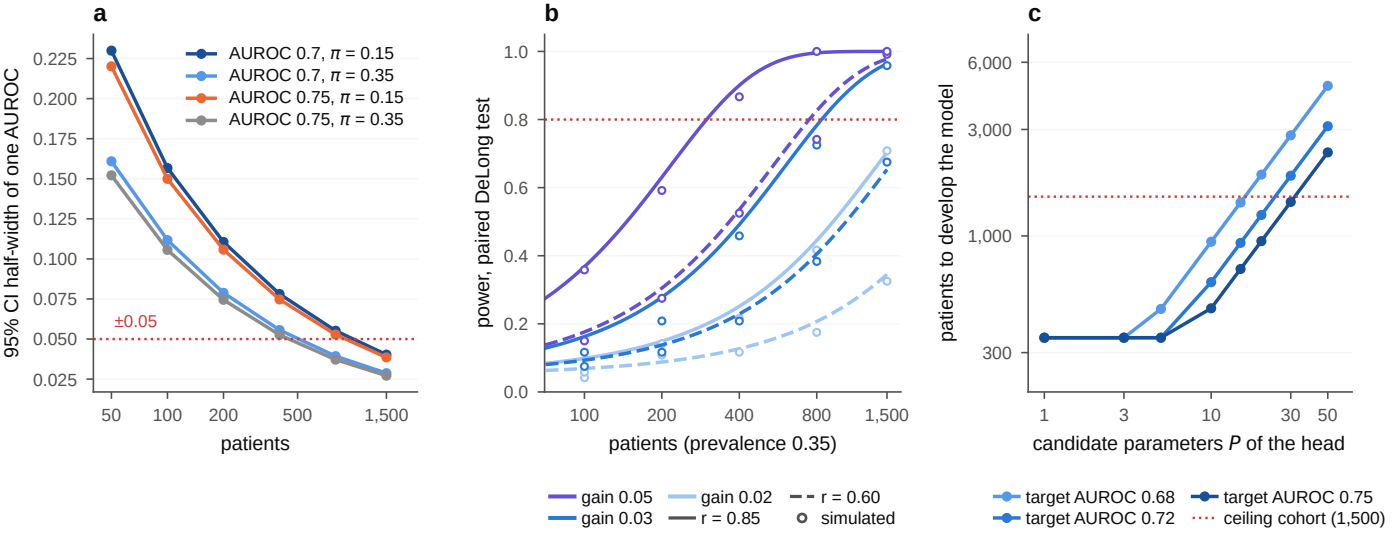
a
AUROC 0.7, π = 0.15
AUROC 0.7, π = 0.35
AUROC 0.75, π = 0.15
AUROC 0.75, π = 0.35
95% CI half-width of one AUROC
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
±0.05
50
100
200
500
1,500
patients
b
power, paired DeLong test
1.0
0.8
0.6
0.4
0.2
0.0
100
200
400
800
1,500
patients (prevalence 0.35)
gain 0.05
gain 0.03
gain 0.02
r = 0.85
r = 0.60
simulated
c
patients to develop the model
6,000
3,000
1,000
300
1
3
10
30
50
candidate parameters P of the head
target AUROC 0.68
target AUROC 0.72
target AUROC 0.75
ceiling cohort (1,500)